# Day 4: Other Protocols in Action (FTP and SNMP)

**Theory: FTP and SNMP**

**FTP (File Transfer Protocol):** FTP is used to upload and download files between a client and a server over a network. It uses TCP port 21 for the control connection (login and commands) and a separate data connection for the file itself (port 20 in active mode, or a passive port range). Normal FTP sends the username, password and data in plain text, so FTPS (FTP over TLS) or SFTP (over SSH) is used when security is needed.

**SNMP (Simple Network Management Protocol):** SNMP is used by network administrators to monitor and manage devices such as routers, switches, access points, servers and printers. A manager (monitoring software) asks an agent (running on the device) for information like CPU load, interface traffic, temperature and uptime. The information is stored in a MIB (Management Information Base) and identified by OIDs. SNMP uses UDP port 161 (queries) and UDP port 162 (traps, which are alerts sent by the device). SNMPv1 and v2c use only a "community string" (like a password) in plain text, while SNMPv3 adds authentication and encryption.

| Point | FTP | SNMP |
|---|---|---|
| Purpose | Transfer files | Monitor and manage network devices |
| Port | TCP 21 (control), 20 or passive range (data) | UDP 161 (queries), UDP 162 (traps) |
| Works between | FTP client and FTP server | SNMP manager and SNMP agent |

### Task 1. Install FileZilla Server on your Windows machine, create a user account, and set up a shared folder for FTP access.

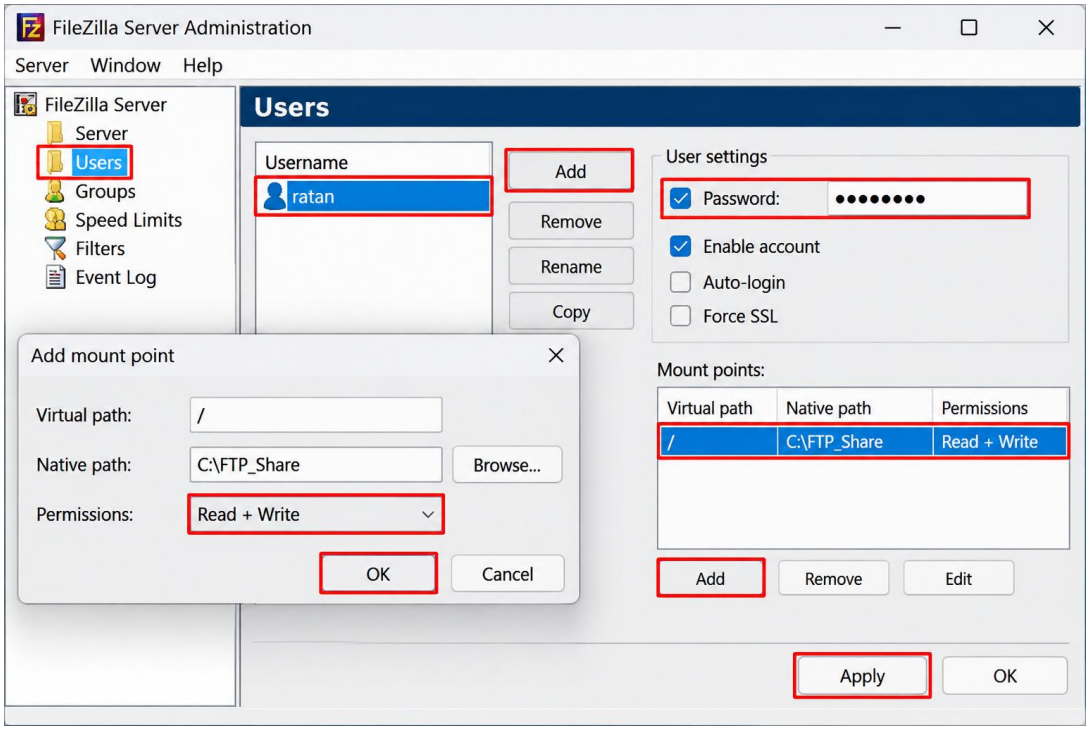
**Answer:**

**Steps:**

1. Download FileZilla Server from the official site (filezilla-project.org) and install it with the default options. Keep the "FileZilla Server Administration Interface" selected.
2. Create the shared folder on the computer, for example C:\FTP_Share (create a normal folder with this name).
3. Open the FileZilla Server Administration Interface and connect to the server (localhost, port 14148). Set the administration password when it asks.
4. Go to Server > Manage users and click Add. Enter a user name (for example ratan) and set a password.
5. In the Mount points section of this user, set Virtual path = / and Native path = C:\FTP_Share. Set Permissions to Read + Write (allow files and directories to be created, changed and deleted). Click OK / Apply.
6. Check that the FileZilla Server service is running (green status) and that port 21 is used. If Windows Firewall asks, allow FileZilla Server.

| Setting | Value I used |
|---|---|
| FTP user name | ratan (or my own name) |
| Password | (set by me, not written here) |

| Setting | Value I used |
|---|---|
| Shared folder (native path) | C:\FTP_Share |
| Virtual path | / |
| Permissions | Read and Write |
| Server address / port | 127.0.0.1 (localhost) / 21 |



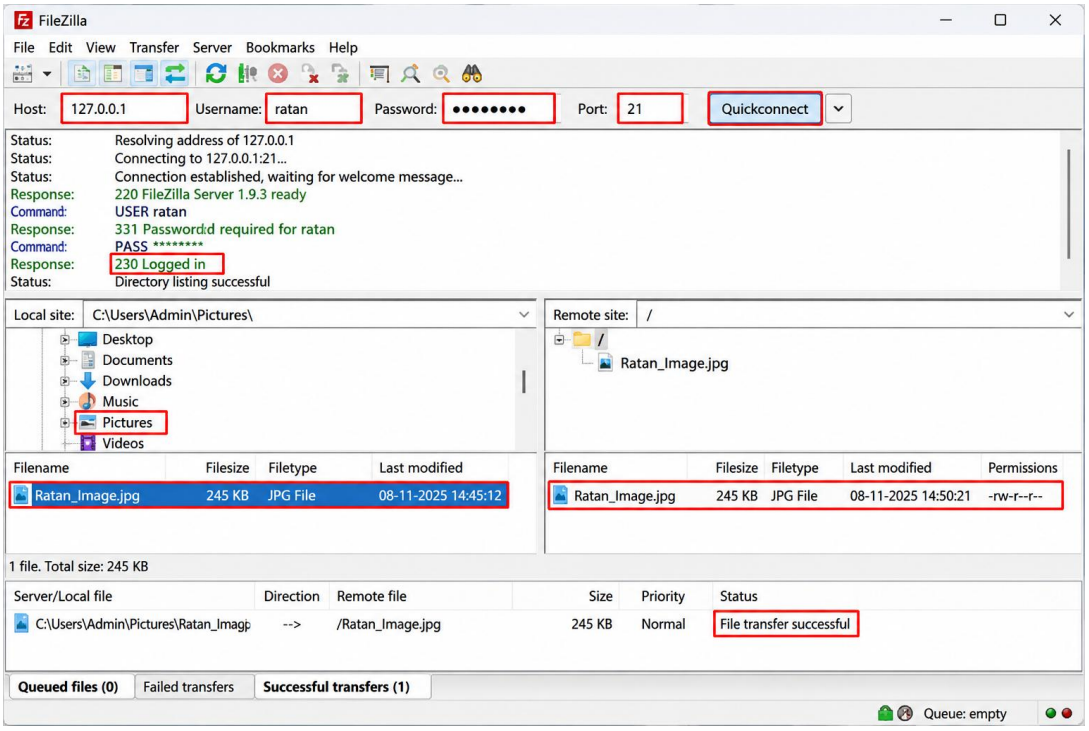

*Screenshot: FileZilla Server Administration > Users: user ratan with password, and the mount point / = C:\FTP_Share with Read + Write permission*

**Task 2. Use FileZilla Client to connect to your FTP server using the credentials you created, and upload any image file from your computer to the shared folder.**

**Answer:**

1. Install and open FileZilla Client.
2. In the Quickconnect bar enter: Host = 127.0.0.1 (or the IP address of the computer), Username = the user created in Task 1, Password = the password, Port = 21. Click Quickconnect.
3. FileZilla Server 1.x uses FTP over TLS, so the client shows an "Unknown certificate" box. Tick "Always trust this certificate" and click OK (it is my own server).
4. The status log shows "Logged in" and the right side (Remote site) shows the shared folder.
5. On the left side (Local site) open the folder with an image, for example Pictures. Note its file name and size.
6. Drag the image to the Remote site side (or right-click > Upload). The status log shows "File transfer successful".
7. Open C:\FTP_Share in File Explorer to confirm the image is there.

| Item | Value |
|---|---|
| Image file name | Ratan_Image.jpg |
| Original size (local) | 245 KB (250,880 bytes) |
| Server response | 220 FileZilla Server 1.9.3 ready; 230 Logged in; File transfer successful |



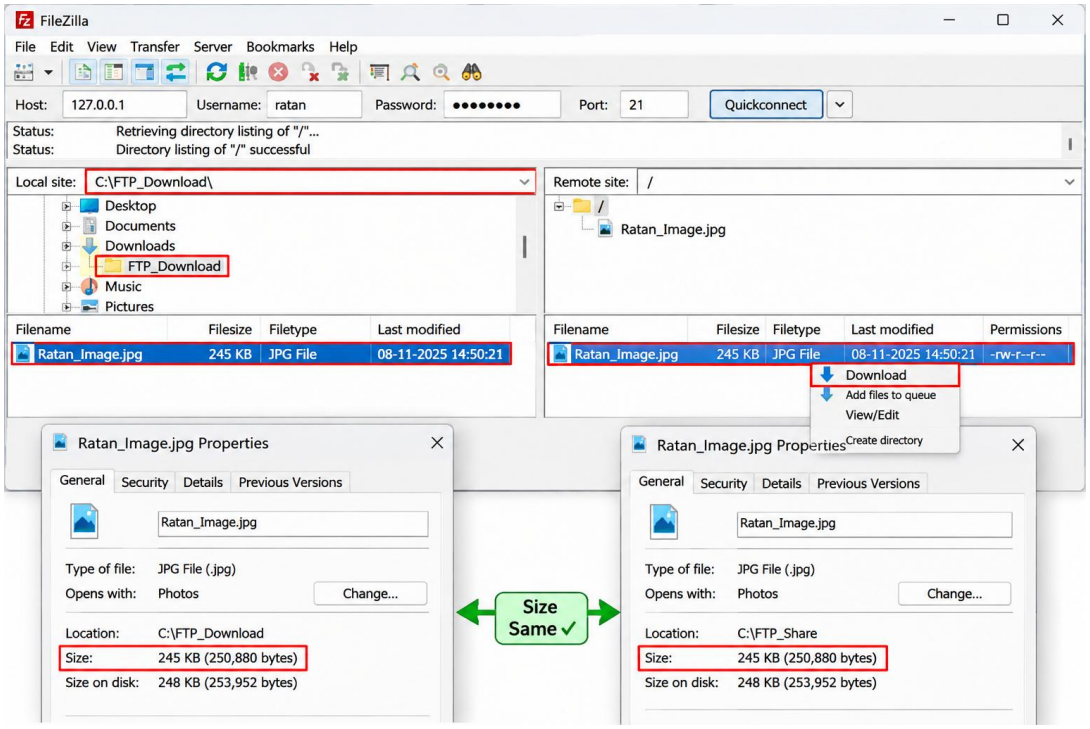

*Screenshot: FileZilla Client connected to 127.0.0.1 (230 Logged in). Ratan_Image.jpg (245 KB) from the Pictures folder is uploaded to the server (File transfer successful)*

## Task 3. Download the uploaded file from your FTP server to a different folder on your computer using FileZilla Client, and verify the file size matches the original.

**Answer:**

1. In FileZilla Client, change the Local site to a different folder, for example C:\FTP_Download (create it first).
2. In the Remote site, right-click the uploaded image and choose Download (or drag it to the Local site).
3. The log shows "File transfer successful".
4. Verify the size: right-click both files (the original and the downloaded one) > Properties, or compare the Filesize column in FileZilla. Both sizes must be exactly the same in bytes.

| | Original file | Downloaded file |
|---|---|---|
| Folder | C:\FTP_Share (server folder) | C:\FTP_Download |
| Size (bytes) | 250,880 bytes (245 KB) | 250,880 bytes (245 KB) |
| Same? | Yes, both sizes are the same | |




*Screenshot: Ratan_Image.jpg downloaded to C:\FTP_Download. The Properties of the downloaded file and the file in C:\FTP_Share both show 245 KB (250,880 bytes), so the size matches*

Result: The file was downloaded from the FTP server to a different folder (C:\FTP_Download). Both files have the same size, 250,880 bytes, so the file was transferred completely without any change.

### Task 4. Research and list 3 real-world scenarios where SNMP is used in network management for apps or services you use (such as ISPs, campus WiFi, or large websites).

**Answer:**

| No. | Scenario | How SNMP is used | What the admin gets |
|---|---|---|---|
| 1 | ISP (internet service provider, for example a broadband or mobile company) | The monitoring system polls the routers and switches of the ISP regularly and reads interface counters (bytes in and out) of every link. | Bandwidth usage graphs, detection of overloaded links, and alerts when a link goes down, so the ISP can fix it before many customers complain. |
| 2 | Campus or office Wi-Fi (for example a college network) | The Wi-Fi controller and the access points act as SNMP agents. The management software reads the number of connected users, signal and load of each access point, and receives traps when one stops working. | Admins see which access point is overloaded or offline (for example in the library) and can repair or add more access points. |
| 3 | Large websites and data centers (for example a shopping, food-delivery or streaming service) | The servers, load balancers and switches in the data center send CPU, memory, disk, temperature and uptime values to a monitoring tool (such as Zabbix, Nagios or PRTG) through SNMP. | Early warning before a server fails or becomes too hot or slow, so the website stays available during heavy traffic. |

In all three cases the idea is the same: the manager asks the devices for their health, bandwidth and uptime values (SNMP GET, using UDP port 161), and the devices send an alert (SNMP trap, UDP port 162) when something goes wrong. Using SNMPv3 makes this safer because it encrypts and authenticates the messages.

### Task 5. Create a table comparing FTP and HTTP across at least 5 points: port number, security, data transfer method, use cases, and authentication.

**Answer:**

| Point | FTP (File Transfer Protocol) | HTTP (HyperText Transfer Protocol) |
|---|---|---|
| Port number | TCP 21 for control (commands, login) and TCP 20 (active mode) or a passive port range for data | TCP 80 (HTTP) and TCP 443 (HTTPS) |
| Security | Plain FTP is not secure: user name, password and files go in clear text. Security is added with FTPS (FTP over TLS) or SFTP (over SSH). | Plain HTTP is not secure (clear text). HTTPS adds TLS encryption and is used by almost all websites today. |
| Data transfer method | Uses two connections: a control connection and a separate data connection. It is stateful (the server remembers the session). Made for transferring whole files, also supports upload, download, rename and delete. | Uses a request and response model over one connection. It is stateless (each request is separate). Mainly used to download web pages and data; upload is done with methods like POST and PUT. |
| Use cases | Uploading website files to a hosting server, large file transfers, backups, sharing files between devices on a network. | Browsing websites, web applications, REST APIs, streaming and downloading content in the browser. |
| Authentication | Normally requires a user name and password (anonymous login is also possible). Permissions are given per user. | Not required by default (public pages). When needed it uses Basic or Digest authentication, cookies and session tokens, or login forms. |
| Client software | FTP client such as FileZilla, WinSCP or the command line | Web browser, curl, Postman or an app |